

When Is an Inverse a Function?

TEACHER KEY | Algebra II | Unit 1 | Day 12 | TEK A2.2.C

Objective: I can decide whether the inverse of a function is itself a function, and restrict the domain of the original when it is not.

Part 1 — The Problem with $f(x) = x^2$

Fill in the table, then look for repeats in the bottom row.

x	-3	-2	-1	0	1	2	3
$f(x) = x^2$	9	4	1	0	1	4	9

Which outputs appear more than once? **1, 4, and 9** So f is **not** one-to-one.

Now swap to build the inverse. Both (1, 1) and (1, -1) appear. Is that a function? **no**

Why not? **the input 1 would have two different outputs**

Written out, the inverse relation is $y = \pm\sqrt{x}$, and the $\pm$ is exactly the problem.

Part 2 — Fix It with a Restriction

Keep only half of the parabola, so no output repeats.

Restriction on $f(x) = x^2$	Which half of the parabola	Then $f^{-1}(x) =$
$x \geq 0$	the right half	$+\sqrt{x}$
$x \leq 0$	the left half	$-\sqrt{x}$

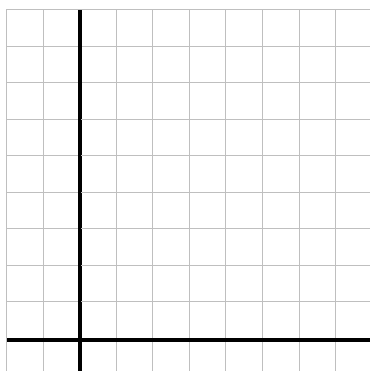
Either restriction works. The usual choice is **$x \geq 0$** , which gives the familiar $f^{-1}(x) = \sqrt{x}$.

Restricting the domain of f also restricts the **range** of f^{-1} .

Part 3 — Graph the Restricted Pair

Each square is 1 unit. Graph only the half of x^2 you kept, then its inverse, plus $y = x$.

$f(x) = x^2$ for $x \geq 0$, and $f^{-1}(x) = \sqrt{x}$



Record it.

Domain of restricted f : **$[0, \infty)$**

Range of restricted f : **$[0, \infty)$**

Domain of f^{-1} : **$[0, \infty)$**

Do they mirror across $y = x$? **yes**

Part 4 — Which Pairs Need No Restriction?

Some parent functions are already one-to-one, so their inverses are functions with no help.

Function	One-to-one?	Restriction needed?	Its inverse
$f(x) = x^2$	no	yes	$\sqrt{x}$
$f(x) = x^3$	yes	no	$\sqrt[3]{x}$
$f(x) = 2^x$	yes	no	$\log_2(x)$
$f(x) = 2x + 1$	yes	no	$(x - 1)/2$
$f(x) = x $	no	yes	$\pm x$

Pattern: functions that only **increase** (or only decrease) are one-to-one. Ones that turn around, like x^2 and $|x|$, are not.

Part 5 — Quick Check

1. Is the inverse of every function also a function? **no**
2. Which test tells you the inverse will be a function? **the horizontal line test**
3. Why does x^3 need no restriction but x^2 does? **x^3 never repeats an output; x^2 does**

Practice

Decide whether a restriction is needed, and give one if so.

1. Does $f(x) = |x|$ have an inverse that is a function with no restriction? **no**
2. Restrict $f(x) = |x|$ so its inverse is a function. Give the restriction and the inverse.
 $x \geq 0$, giving $f^{-1}(x) = x$
3. Is $f(x) = x$ to the 4th power one-to-one? **no** Give a restriction that fixes it. **$x \geq 0$**
4. True or false: $f(x) = x^3 + 1$ needs a domain restriction for its inverse to be a function. **false - it is already one-to-one**
5. Explain, using the horizontal line test, why $f(x) = x^2 - 4$ has an inverse that is not a function.

6. Give one restriction (there are two possible) that makes $f(x) = x^2 - 4$ a one-to-one function. **$x \geq 0$ (or $x \leq 0$)**

Exit Ticket

Consider $f(x) = x^2 - 4$ with no restriction.

Is its inverse a function? **no** Give a restriction that fixes it: **$x \geq 0$ (or $x \leq 0$)**

Explain in one sentence why the restriction is needed: **without it two inputs share an output, so the inverse fails the vertical line test**